Quantification: 536-63x_Hyp_RD_z03c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	24
interior lacuna count	20
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 20)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2246	2189	1349.36	px^2
major_axis_length_px	lacuna shape	91.1698	77.2965	50.4613	px
minor_axis_length_px	lacuna shape	36.6156	35.0225	10.5517	px
aspect_ratio	lacuna shape	2.4415	2.0058	1.2094	unitless
eccentricity	lacuna shape	0.8504	0.8669	0.1079	unitless
solidity	lacuna shape	0.7616	0.7697	0.1192	unitless
roots_count	headline	9.9	8	7.3692	count
ring_length_r30_px	headline	277.95	227	189.242	px
ring_length_r60_px	reported	533.15	501.5	372.039	px
owned_length_px	ownership-dependent	561.17	443.549	603.473	px
edge_count	ownership-dependent	28.8	23	30.9084	count
mean_edge_length_px	ownership-dependent	20.1922	20.2443	6.6628	px
perimeter_px	normalised	288.988	236.072	176.386	px
ring_area_r30_px2	normalised	8449.85	7402	4330.81	px^2
ring_area_r60_px2	normalised	18872.5	17524.5	10079.4	px^2
in_frame_fraction_r30	normalised	0.9814	1	0.0631	unitless
in_frame_fraction_r60	normalised	0.967	1	0.0934	unitless
ring_density_r30	normalised	0.0347073	0.0328068	0.0163136	px^-1
ring_density_r60	normalised	0.0304962	0.0269805	0.0155834	px^-1
roots_per_100px_perimeter	normalised	3.637	3.4813	1.9035	per 100 px
ring_attached_length_r30_px	network v2	245.85	211	187.029	px
ring_attached_length_r60_px	network v2	395.2	321.5	344.693	px
ring_length_w_r30_px	network v2	314.918	262.38	217.107	px
ring_length_w_r60_px	network v2	603.035	573.952	424.859	px
sholl_crossings_r10	network v2	7.2	6.5	5.0325	count
sholl_crossings_r20	network v2	6.9	7	4.2661	count
sholl_crossings_r30	network v2	6.95	7	5.1142	count
loop_count	loops, report only	0.95	0	1.7614	count
loop_edge_share	loops, report only	0.0725	0	0.1074	share
ring_width_mean_r30_px	width	7.2791	7.208	0.6944	px

Per field
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measure	value	units
analysed_area_px2	980857	px^2
total_skeleton_length_px	21555	px
canalicular_length_density_per_px	0.0219757	px^-1
junction_count	396	count
junction_density_per_px2	0.00040373	px^-2
lacuna_count	24	count

lacunae_per_px2	2.447e-05	px^-2
skeleton_component_count	390	count
mean_component_length_px	55.2692	px
median_component_length_px	20	px
field_length_density_w_per_px	0.0248332	px^-1
field_density_without_flagged_per_px	0.0223511	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	940	px
unattached_component_count	51	count
canalicular_length_density_attached_per_px	0.0219757	px^-1
field_length_density_with_unattached_per_px	0.022934	px^-1
loop_count	32	count
loop_edge_share	0.1014	share
edge_count	1174	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.52	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	21282	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1311	70.1057	29.6291	2.3661	0.9063	0.7627	1.4823	11.1503	890.967	True
2	3585	211.287	33.1642	6.3709	0.9876	0.6063	-0.3562	190.729	656.387	False
3	4424	168.508	61.7794	2.7276	0.9304	0.5066	-0.0985	178.037	485.058	False
4	3341	92.9975	56.4772	1.6466	0.7945	0.7441	-0.7147	129.753	115.198	False
5	740	35.5513	27.759	1.2807	0.6248	0.9091	-0.3246	161.099	832.037	False
6	722	38.3774	27.8046	1.3803	0.6893	0.7697	-1.3522	186.446	142.362	False
7	1683	63.5962	39.0607	1.6281	0.7892	0.7571	-0.4294	213.583	88.5187	False
8	2785	110.226	41.3878	2.6632	0.9268	0.6921	-0.2074	250.038	115.956	False
9	1189	59.3046	41.9925	1.4123	0.7061	0.5118	0.1219	253.995	38.2666	False
10	3218	123.087	35.0225	3.5145	0.9587	0.8067	-0.0876	285.404	998.443	False
11	4546	152.562	46.2815	3.2964	0.9529	0.6884	-0.3441	401.046	256.605	False
12	408	31.1169	21.0378	1.4791	0.7368	0.7612	-0.0706	397.571	142.706	False
13	3934	133.278	41.5402	3.2084	0.9502	0.7505	-0.3234	452.159	836.983	False
14	2906	117.901	38.3023	3.0782	0.9458	0.7758	-0.3279	473.994	376.288	False
15	2475	101.315	32.9688	3.0731	0.9456	0.8624	-0.3801	520.411	501.561	False
16	5784	165.174	57.0067	2.8975	0.9386	0.8298	-0.2128	680.673	1000.68	True
17	8146	159.748	75.4405	2.1175	0.8815	0.8502	0.1567	804.35	989.374	True
18	2189	73.1101	40.4728	1.8064	0.8328	0.9132	-0.5214	637.506	936.232	False
19	1176	57.7542	28.7935	2.0058	0.8669	0.8276	-0.2231	683.004	402.083	False
20	679	37.9692	24.2376	1.5665	0.7697	0.8573	-0.1349	734.916	174.795	False
21	829	46.9885	25.6067	1.835	0.8385	0.8399	0.1493	873.288	867.089	False
22	9467	148.982	117.161	1.2716	0.6177	0.7637	0.4415	974.316	939.645	True
23	1845	77.2965	32.007	2.415	0.9102	0.8909	0.0147	929.902	369.506	False
24								60.2404	102.908	False

Network per lacuna

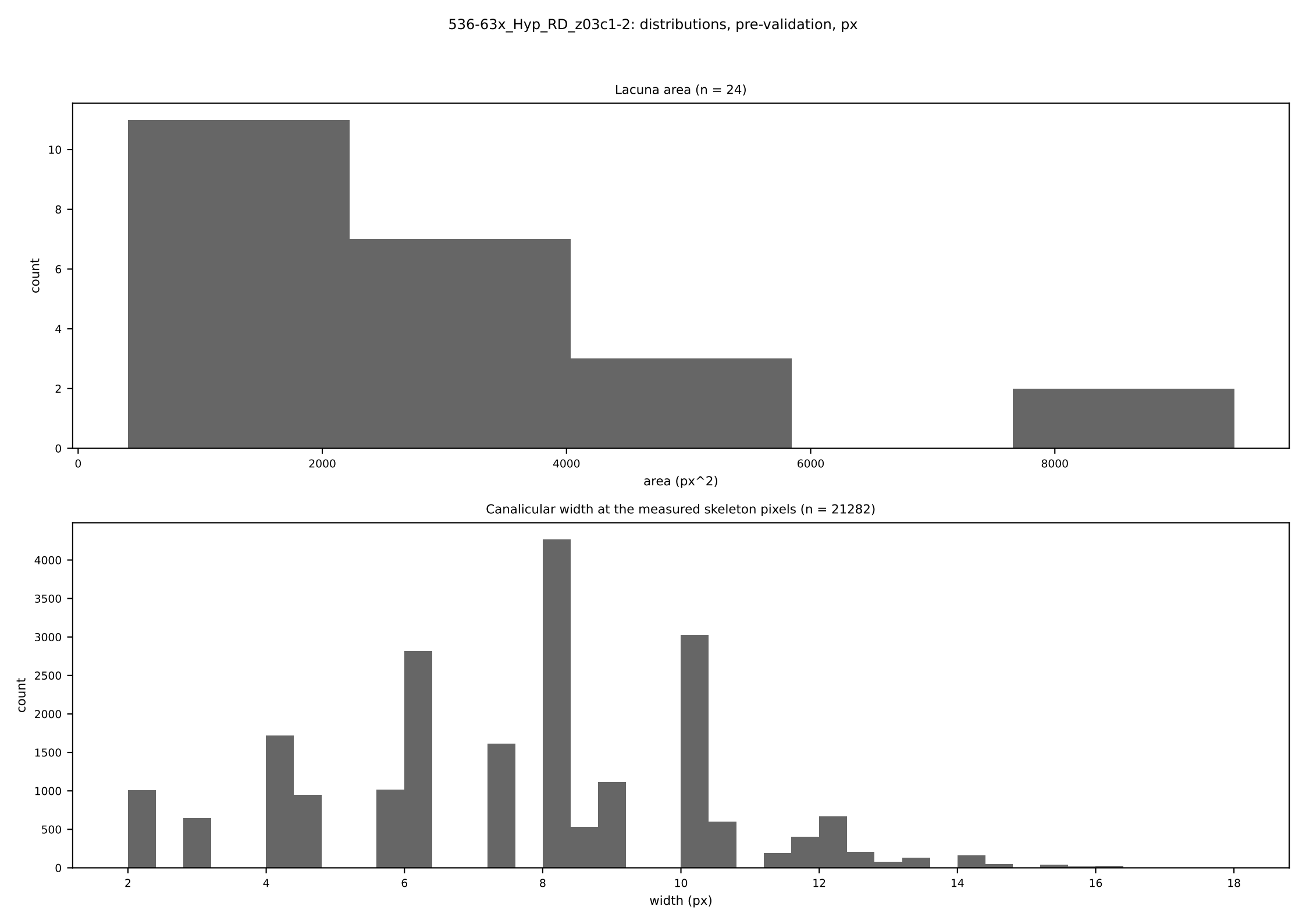
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	7	241	470	584.032	24	24.3346	7.4473
2	25	436	1059	1229.76	69	17.8226	7.8048
3	31	909	1712	2780.85	143	19.4465	7.2606
4	10	334	568	438.321	24	18.2634	7.9773
5	7	189	377	181.983	12	15.1652	8.8876
6	2	123	144	64.2843	3	21.4281	6.5944
7	8	128	142	207.48	17	12.2047	6.5266
8	8	242	400	448.777	24	18.699	6.9291
9	15	405	487	381.421	29	13.1525	6.497
10	7	184	421	472.676	20	23.6338	7.1509
11	15	432	828	570.931	26	21.9589	7.3319
12	8	333	648	928.896	40	23.2224	7.9893
13	13	383	716	707.21	25	28.2884	7.0866
14	11	313	599	309.492	15	20.6328	7.9034
15	9	212	518	348.22	17	20.4835	7.7662
16	2	136	167	86.6985	5	17.3397	
17	4	80	113	50.4558	4	12.614	7.3492
18	2	60	67	28.8995	3	9.6332	
19	9	273	543	479.86	31	15.4794	7.208
20	8	202	508	486.475	22	22.1125	7.5785
21	1	44	71	41.7279	1	41.7279	5.8562
22	0	28	53	0	0	0	4.3619
23	4	145	360	302.037	15	20.1358	6.7922
24	5	212	495	814.11	40	20.3527	7.1629

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	201.782	4986	12541	0.6061	0.574	0.0483353	0.0374771	3.4691
2	655.855	17276	38843	1	1	0.0252373	0.0272636	3.8118
3	733.068	18106	38869	1	1	0.0502044	0.0440454	4.2288
4	398.031	8060	14497	1	1	0.0414392	0.0391805	2.5124
5	114.811	5937	17499	1	1	0.0318343	0.0215441	6.097
6	164.652	3157	5850	1	1	0.038961	0.0246154	1.2147
7	225.593	2244	2366	1	1	0.057041	0.0600169	3.5462
8	361.789	7027	12429	1	1	0.0344386	0.0321828	2.2112
9	298.433	6199	9201	0.8958	0.756	0.0653331	0.052929	5.0262
10	328.049	8327	18178	0.7322	0.645	0.0220968	0.0231599	2.1338
11	472.659	14091	29855	1	1	0.0306579	0.027734	3.1735
12	110.154	5584	12776	1	1	0.0596347	0.0507201	7.2625
13	380.534	12648	30304	1	1	0.0302815	0.0236272	3.4163
14	324.635	11576	25943	1	1	0.0270387	0.0230891	3.3884
15	246.551	9801	24158	1	1	0.0216304	0.0214422	3.6504
16	460.794	5167	7599	0.5692	0.5578	0.0263209	0.0219766	0.434
17	525.623	8185	11797	0.6487	0.6114	0.00977398	0.00957871	0.761
18	224.693	7360	14921	1	0.995	0.00815217	0.00449032	0.8901
19	188.93	7444	20339	1	1	0.0366738	0.0266975	4.7637
20	120.125	5980	17550	1	1	0.0337793	0.0289459	6.6597
21	144.368	5916	13005	1	1	0.00743746	0.00545944	0.6927
22	703.76	10275	13592	0.6197	0.5585	0.00272506	0.00389935	0
23	204.995	8412	22105	1	0.9866	0.0172373	0.0162859	1.9513
24	81.8406	3852	8762	1	0.9576	0.0550363	0.056494	6.1094

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	183	432	272.894	529.304	3	8	5
2	366	870	490.517	1198.23	15	10	11
3	887	1578	1041.62	1954.8	21	19	24
4	308	456	374.907	640.044	11	8	5
5	120	178	204.225	411.764	3	6	4
6	86	86	137.64	161.853	7	2	5
7	128	128	142.811	158.468	2	3	0
8	212	305	275.794	456.818	7	7	7
9	366	430	459.848	547.789	10	5	5
10	180	291	201.368	466.291	5	4	5
11	406	597	485.304	922.909	13	12	12
12	303	472	373.037	735.257	7	10	9
13	358	543	435.889	806.796	10	9	8
14	215	272	364.676	685.269	5	10	9
15	210	338	238.823	587.943	6	7	7
16	108	108	145.74	176.225	3	5	1
17	47	47	85.9411	127.225	4	0	2
18	32	32	63.0416	70.8701	1	1	0
19	233	371	315.907	635.441	8	8	7
20	123	247	233.894	572.831	3	6	8
21	40	49	47.799	76.1127	1	1	2
22	0	0	27.3137	54.4558	0	1	4
23	140	218	162.297	396.149	4	3	4
24	204	443	248.965	575.073	5	7	7


536-63x_Hyp_RD_z03c1-2: distributions, pre-validation, px
Lacuna area (n = 24)
count
area (px^2)
0
2000
4000
6000
8000
0
2
4
6
8
10
Canalicular width at the measured skeleton pixels (n = 21282)
count
width (px)
2
4
6
8
10
12
14
16
18
0
500
1000
1500
2000
2500
3000
3500
4000